

[ACJC/2024/Q3]

3 Name your Jupyter Notebook as:

TASK3_<your name>_<centre number>_<index number>.ipynb

For each of the sub-tasks, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 3.1
        Program code
        Output:
```

Task 3.1

A linked list is implemented using Object-Oriented Programming (OOP) to store data about students' names and scores.

Write the `Node` class, including the following attributes:

- `data` stores a tuple of the form `(name, score)`, e.g. `("Sam", 95)`.
- `pointer` points to the following `Node` in the linked list if one exists, or `None` otherwise.

Write the `LinkedList` class, including the following attribute:

- `headpointer` points to the head node of the linked list;

and the following methods:

- `insert_head(data)` inserts a node with the given data at the head of the linked list;
- `display()` displays the data in all the nodes of the linked list in order of the list;
- `search(name)` returns `True` if `name` is present in the data of one of the nodes in the linked list, and `False` otherwise;
- `remove(name)` attempts to delete the node with `name` from the linked list, and returns `True` if the node was successfully deleted, and `False` if `name` does not exist in any node in the linked list;
- `reverse_list()` that reverses the order of the nodes in the linked list. [10]

Task 3.2

Write program code to:

- create an instance of `LinkedList`;
- store the data found in the text file `names.txt` in the `LinkedList`, where the data in each line in the file should be stored in a `Node`;
- call the `reverse_list()` and `display()` methods on the `LinkedList` object. [3]

[ACJC/2024/Q3 Cont']

Task 3.3

Linked lists can be used to resolve collisions in a hash table.

For this question, the data stored would be tuples of the form `(name, score)`, e.g. `("Sam", 95)`. Each tuple is stored inside a `Node` which is inside a `LinkedList`. The hash table consists of an array of `LinkedLists`.

Write the `HashTable` class, including the following attributes:

- `max_size` is the maximum number of items stored in the hash table, initialized to 97;
- `table` is a 1-dimensional array of 97 instances of a `LinkedList`;

and the following methods:

- `hash_name(name)` is a hash function that adds up the ASCII values of the characters in `name` and returns the sum modulo 97 (i.e., the remainder of the sum after dividing by 97);
- `add(tup)` inserts a key-value tuple to the hash table;
- `retrieve(name)` returns the `score` associated with `name` if it is in the hash table, and `False` otherwise;
- `show()` displays all the data stored in the hash table.

In the event of a collision, the `Node` containing the data would be inserted at the head of the `LinkedList` found at the appropriate bucket of the hash table. [8]

Task 3.4

Write program code to:

- create an instance of `HashTable`;
- store the data found in the text file `names.txt` in the `HashTable` object;
- call the `show()` method on the `HashTable` object. [3]

Save your Jupyter Notebook for Task 3.